Neural Memory and Interval Calibration

Chengxuan YUAN Student ID: 50025581
DSAA 2012, HKUST(GZ)

1 Motivation and research question

Volatility shocks challenge calibrated financial intervals. Neural control, score–threshold adaptation, and retrospective updates already address adaptation [1–3]. I ask: **how do current neural scale response and historical score memory jointly affect calibration at shock entry and recovery?** I will investigate dependence on fitted model, training data, and shock duration.

2 Method and evaluation

GRU and causal TCN models learn Gaussian scales from raw past returns, with validation early stopping. At zero center, current context a and archive context b give half-width

$$h_t^{a,b,W} = s_t^a \, Q_{1-\alpha}\big(\{|y_i|/s_i^b\}_{i=t-W}^{t-1}\big), \qquad \alpha = 0.1.$$

Cross $a, b \in \{16, 64\}$ and score windows $W \in \{32, 128\}$, sharing weights within fits; $a = b$ is ordinary calibration. Paired shock/no-shock simulations vary duration, magnitude, tails, and stochastic volatility. Decompose event-minus-control context-by-window contrasts in both current/archive orders. Repeat across five paired seeds and independent training datasets. Controls include EWMA, fitted GARCH, and rolling/adaptive/strongly adaptive calibration [4].

Use generated data and daily log returns from public ECB EUR-cross rates only. The ECB pilot trains through 2018, validates in 2019–2020, and evaluates 2021 onward exploratorily. Report interval score, coverage, width, and simulated oracle deficits; date-block comparisons retain cross-asset dependence. Hyperparameters use validation data; confirmatory simulations use frozen rules and new paths.

3 Expected contribution and feasibility

Expected outcomes are a reproducible current/archive decomposition and a phase-specific explanation of *neural fitting variability*. Exploration and a three-dataset replication demonstrate local feasibility and mixed results. Through November, I will consolidate lag/pulse diagnostics, literature positioning, robustness checks, and public financial analysis, then finalize the report and presentation before December 11.

References

[1] R. Li and A. Rodríguez. Neural Conformal Control for Time Series Forecasting. *AAAI*, 2025.

[2] R. Tumu, A. Singh, and R. Mangharam. AdaptNC: Adaptive Nonconformity Scores for Conformal Prediction under Distribution Shift. *arXiv:2602.01629v2*, 2026.

[3] J. Jun and I. Ohn. Online conformal inference with retrospective adjustment for faster adaptation to distribution shift. *Pattern Recognition*, 180:114406, 2026.

[4] A. Bhatnagar, H. Wang, C. Xiong, and Y. Bai. Improved Online Conformal Prediction via Strongly Adaptive Online Learning. *ICML*, 2023.

Preprint.